

## Secondary examination 2016

### SCIENCE

Time: 3 1/4 hours

Total marks: 80

General instructions for the examinees.

1. Candidates must write first his or her roll number on the question paper compulsorily.
2. All questions are compulsory.
3. Write the Answer to each question in the given Answer book only.
4. For questions having more than one part the Answers to the other part should be written together in continuity.
5. If there is any error /difference /contradiction in Hindi and English versions of the paper the question of Hindi version should be treated valid.

| 6. Section | q.no  | marks |
|------------|-------|-------|
| A          | 1-8   | 1     |
| B          | 9-14  | 2     |
| C          | 15-24 | 3     |
| D          | 25-30 | 5     |

7. There are internal choices in question 28 to 30.

### PART (A)

1. Write the two components of baking powder. [1]

Answer: Sodium bicarbonate and monocalcium phosphate.

2. Write the name of reproductive process in the plants which have lost the capacity to produce seeds. [1]

Answer: Vegetative propagation.

3. Write the atomic weight of chlorine isotopes. [1]

Answer: The atomic weight of chlorine isotopes is 35.5

4. In the following strong acid and strong alkane base is

a) acetic Acid or hydrochloric acid

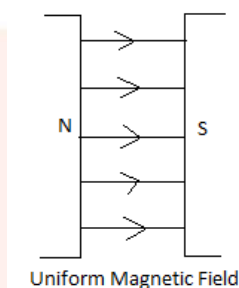
b) Sodium Hydroxide or calcium hydroxide

[1/2 + 1/2=1]

**Answer:** strong acid is hydrochloric acid and strong base is Sodium Hydroxide.

**5. The magnetic field in a given region is uniform. Draw a diagram to represent it.**  
[1]

**Answer:**



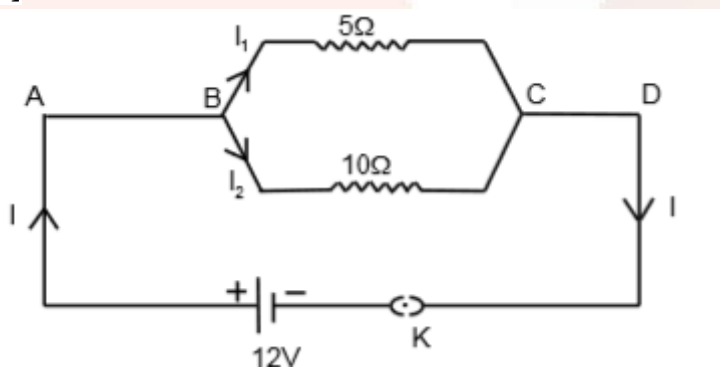
**6. Write the name of acidic oxides which are responsible for acid rain.** [1]

**Answer:** Sulphur dioxide or Nitrogen Oxide.

**7. A current through a horizontal power line flows east to west Direction. What is the direction of magnetic field at a point directly below it.**  
[1]

**Answer:** The direction of magnetic field at a point is towards north direction.

**8. In the given circuit diagram calculate the current  $I_2$  through the resistor of  $10\Omega$ .**  
[1]



$$R_{eq} = \frac{1}{5} + \frac{1}{10}$$

$$= \frac{3}{10}$$

$$= \frac{10}{3}$$

$$I = R/V$$

$$= 12/10 \times 3$$

$$= 3.6$$

$$I_1 \times 5 = I_2 \times 10$$

$$I_1 = 2 \times I_2$$

$$I = I_1 + I_2$$

$$I = 3.6$$

$$3 I_2 = 3.6$$

$$I_2 = 1.2 \text{ A}$$

### PART(B)

**9. What is combination reaction write combination reaction of quicklime with water.**  
**1+1=2**

**Answer:** Reaction in which two or more reactants combine to form a single product is called combination reaction. quick lime reacts with water to form calcium hydroxide called slaked lime. This reaction is highly exothermic so a lot of heat is produced during the reaction.



**10. Write four characteristics of the good source of energy.**  $\frac{1}{4} \times 4 = 2$

**Answer:** A good source of energy can be characterized by following factors-

- It should do large amount of work from the same volume for mass
- It should be easily accessible.
- It should be easy to store and transport from place of their production to consumer.
- It should be economical. It should give more amount of energy at lesser cost.

**11. At night which type of beam of light is used by drivers and why explain it.**  $\frac{1}{2} + 1 \frac{1}{2} = 2$

**Answer:** Concave Mirrors are used to obtain parallel beam of light in the vehicle. It helps the driver to see far objects. In busy traffic system the light beams come from front side dazzle the driver hence in busy traffic or at night driver should use low intensity light beams. High intensity

light should be used at the national highways so that the driver can see far object from a proper distance.

**12. Describe homologous organs with the help of example. [  $\frac{1}{2} + 1^{\frac{1}{2}} = 2$  ]**

**Answer:** Homologous organs are similar in basic structure and have the same development origin. They have different functions in different organisms. Example the forelimbs of vertebrates (lizard, man, frog and bird) have similar structure but perform different functions.

**13. Why do we have two eyes for vision and not just one. Explain the fact on the basis of**

**a) Field of view [1]**

**b) Dimensional view [1]**

**Answer:** A human eye has a horizontal field of view of about  $150^\circ$  with one eye and  $180^\circ$  with two eyes. So, two eyes give a wider field of view. The ability to detect faint objects is enhanced with 2 eyes.

We have two eyes that are slightly apart, giving us slightly different vantage point. Our brain then processes the 2D images from each eye and gives us a 3D image.

**14. Write the definition of fossils and explain the process of dating of fossil. [1 + 1 = 2]**

**Answer:** Fossil is the naturally preserved remains or traces of animals or plants that lived in the geological past. Fossils are formed when minerals in groundwater replace material in bones and tissues creating a replica in stone of the original organism or of its trace.

Radioactive elements or isotopes are the atoms which release radiations with time and this causes a decrease in the amount and convert to other isotope. This is known as radioactive decay. Half-life of a radioactive element is the time after which the amount of element remains half. To accurately date fossils one should know the half-life of an element. In radioactive decay one isotope changes to another by emitting radiation with time. We check that amount of secondary isotope and then relate it to half-life of the elements to know the time that how much time it takes to make this secondary element.

### PART (C)

**15. a. Write the chemical reaction of the photosynthesis and events occur during the process? [2+1=3]**

**Answer-** photosynthesis involves the following chemical reaction:



Major Events of photosynthesis are followings:

- i) absorption of light energy by chlorophyll
- ii) conversion of light energy into chemical energy and splitting of water molecules into hydrogen and oxygen
- iii) Reduction of carbon dioxide to carbohydrates

**b. which type of mirror is used in headlights of vehicles ?**

**Answer:** Concave Mirrors are used in vehicle headlights to get powerful parallel beam of Light.

**16. a. Which hormone is secreted by adrenal gland? Write any one function of this hormone. [1+1=2]**

**Answer:** Adrenaline is secreted by adrenal gland. Adrenaline helps in defense of the body in emergency situation. It also maintains the correct salt balance in the blood.

**b. Write any two functions of batteries in vehicle. [1]**

**Answer:** Batteries primary function used to start the engine. It is the main source of electricity in the vehicle. Battery has the responsibility to send electricity into the starters to initially start the flywheel .other functions of Battery are blowing of horn, lighting of lamps, etc.

**17. What is Redox reaction in the reaction  $\text{O} + \text{C} = \text{Zn} + \text{CO}$  which substance gets oxidized and which substance gets reduced. [1+1+1=3]**

**Answer:** Those reactions in which oxidation and reduction takes place simultaneously are called Redox reaction in the given reaction Zn is being reduced because it is losing oxygen and carbon is oxidized as oxygen is added to it.

**18a. Write down the formula for magnification obtained by a spherical mirror. [1]**

**Answer:** Magnification is expressed as the ratio of height image to the height of object.

Magnification (m) = height image ( $h_1$ )/height of object ( $h_0$ ).

**b. Name a mirror that can give an erect and enlarged image of an object. [1]**

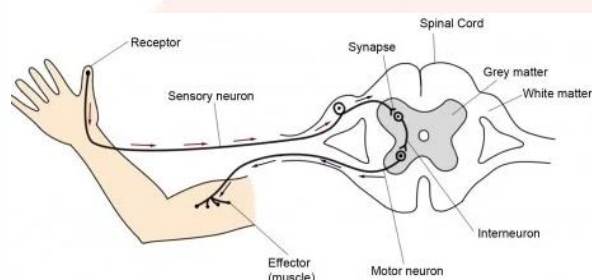
**Answer:** Convex mirror

**c. Define the principal focus of a convex mirror. [1]**

**Answer:** Principal focus is the point on the principal axis of the mirror at which the light rays coming parallel to principal Axis after reflection actually meet or appear to be coming from it is represented by F. In convex mirror the focus is behind the mirror.

**19. Draw a labelled diagram of reflex arc and explain reflex action. [1+2=3]**

**Answer:**



### REFLEX ARC

The pathway taken by nerve impulse in a reflex action is called reflex arc. It is found in the spinal cord although all the information input also goes on to reach the brain.

**20 Match the column X to column Y. [1/2 × 6=3]**

#### Column X (Physical Quantity)

1. Electric current
2. Potential Difference
3. Ohm's Law
4. Resistivity
5. Series combination of resistors
6. Parallel combination of resistors

#### Column Y (Formula)

- a.  $R = R_1 + R_2 + R_3$
- b.  $p = RA/l$
- c.  $1/R = 1/R_1 + 1/R_2$
- d.  $V = W/Q$
- e.  $I = q/t$
- f.  $V = IR$

**Answer:**

|                      |            |
|----------------------|------------|
| Electric current     | $I = q/t$  |
| Potential Difference | $V = W/Q$  |
| Ohm's Law            | $V = IR$   |
| Resistivity          | $p = RA/l$ |

Series combination of resistors

$$R = R_1 + R_2 + R_3$$

Parallel combination of resistors

$$1/R = 1/R_1 + 1/R_2$$

**21. If an egg is not fertilized then what happens explain it. [1+2=3]**

**Answer:** Fertilization is a process of fusion of egg and sperm. Fertilization occurs in the body of female partner. If egg is not fertilized then it means that male sperm cell has not reached ovum. If fertilization does not occur then the egg and the inner lining of uterus along with its blood vessels are shed off. This causes bleeding called menstrual flow.

**22. Write new lands law of octaves. [1+1+1=3]**

**Answer:** John Newland arranged the known elements in order of increasing atomic masses and found that every eighth element had properties similar to that of first. we compare this with the object found in music i.e. Sa, Re, Ga, Ma, Pa, Dha, Ni and in the West the use the notation do, re, MI, FA, so, La, Ti therefore he called it lot of octaves.

OR

**Write the name of element having similar properties of the following.**

- a) Nitrogen
- b) Lithium

**Answer:** Phosphorus has the having similar properties as of Nitrogen and Sodium have the similar properties as of Lithium.

**23. What is pH scale? Explain that pH change for the mouth is cause of tooth decay. [1+2=3]**

**Answer:** PH is used to measure the strength of acid and base. This scale measures hydrogen and concentration present in the solution. Here P is of German word (Potenz) which is indicator of strength and H is if of hydrogen ion. pH of mouth is approximately 6.5 .After taking the food, the bacteria present in the mouth produces acid by the reaction from remaining food attached in the teeth which increases the pH of mouth. If the value of pH is less than 5.5 the enamel of tooth decayed. So, tooth paste should be done after food.

**24. Electric refrigerator rated 600 W operates 8 hours per day. What is the cost of the energy to operate it for 30 days at rupees 4 per kwh . [3]**

$$\text{Power} = 600\text{w}$$

$$T = 8 \text{ hr}$$

$$P = \text{energy/time}$$

$$600 = e/8$$

$$\text{Energy} = 4800w$$

$$\text{Rupees in watt/hour} = 4/1000 = 0.004$$

$$\text{Rupees of } 4800w \text{ in 30 days} = 4800 \times 0.004 \times 30 = 144 \text{ rs.}$$

## PART –D

**25. a. What do you mean by food chain.** [1]

**Answer:** Food chain is a network of living organism in a community through which energy is transferred in the form of food. It describe the relationship of organism about who eats whom.

**b. Explain the food chain in grassland with an example.** [2]

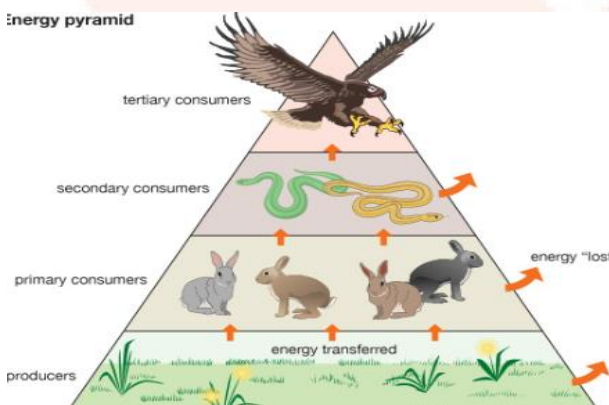
**Answer:** A grassland food chain –

Grass → Insect → frog → Snake → eagle.

The transfer of food or energy takes place through various level in food chains. Grass is a autotroph and producer forms the first level. Grass is eaten by herbivores so it forms the second tropic level which is primary consumer. Herbivores are eaten by snake which forms the secondary consumer. Snake is eaten by eagle which form the tertiary consumer.

**c. Explain the energy flow in an ecosystem with the help of diagram.** [2]

**Answer:** Energy is accumulated by the primary producer and is transferred through food chain in different trophic levels. This phenomenon is called energy flow. It is unidirectional and there is no recycling to previous level.



Energy Flow



**26. a. which metal exists in liquid state at room temperature. [1]**

**Answer:** Mercury exists in liquid state at room temperature.

**b. Write electronic configuration of nitrogen and chlorine atoms. [2]**

**Answer:** Electronic configuration of Nitrogen -2, 5

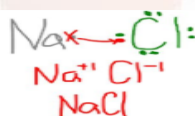
Electronic configuration of Chlorine—2, 8, 7

**c. Show the formation of sodium chloride by the transfer of electrons between nitrogen and chlorine atom. [2]**

**Answer:** If the Sodium and chlorine react with each other than electron loss by Sodium is gained by chlorine and an ionic bond is formed between them.

$\text{Na} \rightarrow \text{Na}^{++} + e^{-}$       2, 8, 1      2, 8

$\text{Cl} + e^{-} \rightarrow \text{Cl}^{-}$       2,8,7      2,8,8



**27. Define alternating current and explain working of the device which produces it. Draw level diagram describe the change of direction of current [1+2+1+1]**

**Answer:** An electric current whose magnitude changes with time and direction reverses periodically is called alternating current. It is noted by AC. Its source are hydroelectric generator, Thermal Power generators, AC generator

Working of AC generator-

It generate the current which changes its direction after equal interval of time which is alternating current. It is a device which is used to transform mechanical energy to electrical energy.

Parts of AC generator

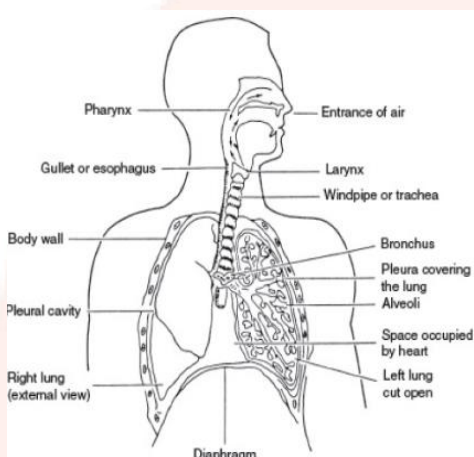
1. Field magnet 2. Armature or coil .Slip ring and brushes.

Working of AC generator

When it starts rotating Let PQ moving up and RS moving down. It cuts the magnetic lines of force now according to Fleming's right hand rule induced current in setup in these arms along the direction of PQ and RS. So effective induced current is obtained flowing in the direction of PQRS. If large number of turns are there in the coil current is produced in each turn that add's up to provide large quantity of current through the rectangular coli. After half cycle of the coil arm RS moves up and arm PQ moves down. Now the direction of the induced current reverses in the coil of in the direction of SRQP so after every half cycle the polarity of current in the arm changes.

**28. a. Draw a labeled diagram of human respiratory system.**

**[2]**



### HUMAN RESPIRATORY SYSTEM

**b. Explain the transportation of Oxygen and Carbon dioxide in human beings.**

**[3]**

**Answer:** During the Transportation in human body the oxygen diffuses across the alveoli and enters the bloodstream. It is transported to the tissue where it is unloaded the carbon dioxide diffuses out of the blood into the alveoli and expel out of the body. The transport of oxygen from the lungs to tissues by blood is due to the ability of hemoglobin to combine reversibly with oxygen.

**OR**

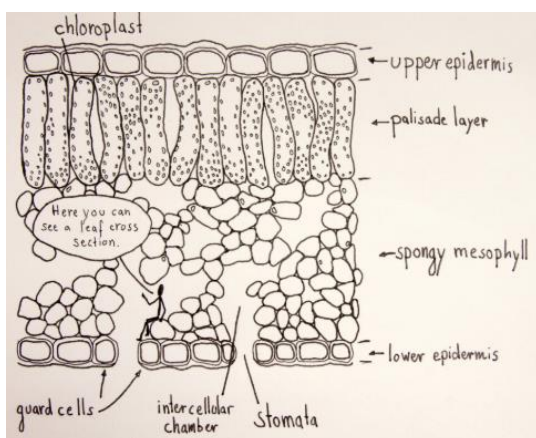
**a) What is respiration?**

**[1]**

**Answer:** Respiration is the oxidation of nutrients in the living cells to release in energy for biological work it takes place in all the cells of the body .Respiration produce carbon dioxide a toxic substance which is eliminated from the body.

b) Draw a labeled diagram of cross section of leaf.

[2]



**Cross Section of Leaf**

c) Explain the transportation of food and other substances in plants.

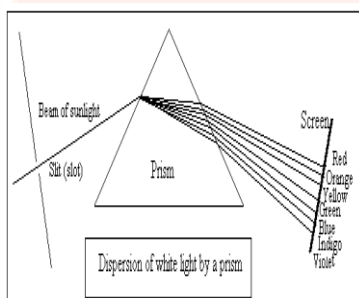
[2]

**Answer:** There is a proper system of transportation in plants. Organized conducting tubes xylem and phloem are present for transportation. Xylem and phloem transport substances from one part of the plant to another. Xylem transports water and Minerals from the soil and phloem transports product of photosynthesis from leaves to other parts of the plants.

29. a. Draw a diagram of refraction of light through a prism and explain the phenomenon of dispersion of light.

[1 + 2 = 3]

**Answer:**



### REFRACTION OF LIGHT THROUGH PRISM

When the light of sun is passing through prism the light splits into seven constituent colors on the screen. This is called dispersion of light. The formation of a band on the screen is called spectrum. The speed of different colors in a medium is different; that's why the spectrum is formed. The sequence of colors is called VIBGYOR. V = Violet, I = indigo, b = blue, G = green, Y = yellow, O = orange, R = red.

**b. Write the formula for lens power and define its unit, [1+1=2]**

**Answer:** The object distance (u), image distance (v), focal length (f) have relation between them which is represented by the following formula in a lens:

$$1/f = 1/v - 1/u$$

Its unit is Dioptre.

**OR**

**A. Explain the following**

**i) Atmospheric refraction and advance Sunrise. [2]**

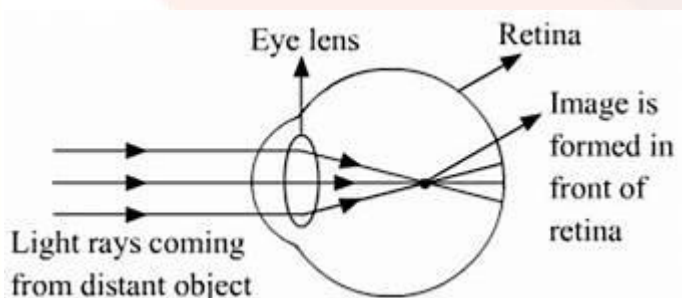
**Answer:** The sun is visible about two minutes before the sunrise. This is because of the atmospheric refraction. When the sun is slightly below the horizon the sunlight coming from less denser medium to more dense air is refracted downward. Because of this the sun appears to be raised Above the Horizon and so the rising of sun can be seen about two minutes before the actual sunrise.

**ii) Tyndall effect. [2]**

**Answer:** The path of a beam of light passing through a true solution is not visible however its path becomes visible when it passes through a colloidal solution where the size of particle is relatively largest. This is scattering of light when it passes through a colloidal solution is called Tyndall effect.

**iii) Draw a ray diagram of an eye suffering from short sightedness. [1]**

**Answer:**



**SHORT SIGHTEDNESS**

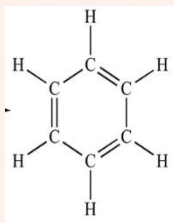
30. a. Write molecular formula of benzene.

[1+2+2]

**Answer:** Molecular formula is  $C_6H_6$

**B. Write structure formula of benzene and write number of triple bonds present in it.**

**Answer:**



**BENZENE**

c. which is Ethane in the following? Write number of cooperative bonds present in it .

- i)  $C_2H_2$  ii)  $C_2H_4$  iii)  $C_2H_6$

**Answer:**  $C_2H_6$

**OR**

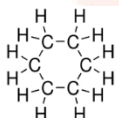
a. Write molecular formula of cyclohexane.

[1]

b. Molecular formula of cyclohexane is  $C_6H_{12}$

c. Draw a structure formula of cyclohexane and write number of cooperative bonds present in it.

**Answer:**



**STRUCTURE OF CYCLOHEXANE**

d. Which is ethene in the following?

[2]

- ii)  $C_2H_2$  ii)  $C_2H_4$  iii)  $C_2H_6$

**Answer:**  $C_2H_4$

**Write number of double bonds present in it.**

**[2]**

**Answer:** One double bond is present

